

Scalability & Future Plan

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

How the current solution can scale to handle more users, more data, or extended features, updated now that the app persists users and prediction history in a database.

Current System Limitations:

S. N O	Limitation	Impact	Priority to Address
1	Manual data entry only; no live weather API	Users must know or look up their own readings	High
2	Single-instance deployment (no load balancing configured)	Limited concurrent user capacity under heavy traffic	Medium
3	No automated test suite	Regressions rely on manual verification	Medium
4	SQLite by default; not ideal for multiple app instances	Prediction history may not persist reliably if scaled beyond one instance	High
5	No password-reset flow or login rate limiting	Locked-out users have no self-service recovery; brute-force login attempts aren't throttled	Medium

Scalability Plan:

S. N O	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Gunicorn process on Render's free tier	Increase Gunicorn worker count; move to a paid Render plan with autoscaling
2	Data Storage (models)	Static Joblib model files bundled with the app	Move model artifacts to object storage (e.g., S3) if models are updated frequently
3	Data Storage (users/history)	SQLite database file on the app instance	Migrate to a managed Postgres database (via DATABASE_URL) for durability across redeployments and multiple instances
4	Performance	Fast (classical ML, sub-second inference)	Add response caching for identical repeated inputs if traffic grows

S. N O	Scalability Aspect	Current State	Proposed Upgrade / Solution
5	Security	Session auth with CSRF protection and hashed passwords; single prediction endpoint per user	Add rate limiting on /login, /register and /predict, plus password-reset email flow, to prevent abuse at scale

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Live weather API integration for automatic input	+1-2 months	Removes manual data entry friction, broadens usability
Phase 3	River/dam water-level data as additional model features	+3-4 months	Improves prediction accuracy with more direct hydrological signals
Phase 4	Time-series forecasting for multi-day lead time	+6 months	Moves from a snapshot classifier to a forward-looking forecast
Phase 5	Password-reset flow, login rate limiting, and a JSON /api/predict endpoint	+1-2 months (parallel with Phase 2)	Hardens the auth layer and opens the prediction engine to external integrations